Personalised Counterspeech using Diffusion Language Modelling

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THESIS CERTIFICATE

This is to certify that the thesis titled **Personalised Counterspeech using Diffusion Language Modelling**, submitted by **Apoorv Jain (2023EET2190)**, to the Indian Institute of Technology, Delhi, for the award of the degree of **Master of Technology**, is a bona fide record of the research work done by him under our supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS: Counterspeech; Diffusion Language Modelling; SEDD; Vector Quantization

With the rising concern over the spread of online hate speech, counterspeech has emerged as a compelling and democratic alternative to content removal. Counterspeech, particularly when delivered by a credible and respected individual, carries greater **persuasive power**, challenging hate speech more effectively and offering stronger support to its victims. Personalized responses can **resonate more with individuals** and also have some credibility with the speaker. This thesis proposes a novel framework for generating personalized and stylistically grounded counterspeech using a **Two-Phase Score Entropy Discrete Diffusion (VQ-SEDD)** model. Unlike traditional autoregressive models, which often lack nuanced and diverse responses, the proposed model leverages the SEDD model with vector quantization to enhance style control and nuance, enabling it to generate persuasive counterspeech in the rhetorical style of credible leaders like **Mahatma Gandhi and Nelson Mandela**.

Experimental results indicate that our method significantly **outperforms generic responses** produced by autoregressive models like GPT, T5, DialoGPT, and LLaMA 3.2 in zero-shot, few-shot, and pretraining settings by **20-30%**, while using **fewer parameters** and yielding comparable or even better performance in some metrics against fine-tuned models. The research highlights the **potential of discrete diffusion models** to topple the era of autoregressive models for controllable text generation tasks. In our study, diffusion models **trained from scratch** with mask denoising objective produced comparable performance to models that were extensively pre-trained with a next-word prediction objective.

By highlighting the significance of the speaker's credibility, this work argues that personalized counterspeech can strengthen both the trustworthiness and effectiveness of online interventions, fostering healthier digital communities while upholding freedom of expression.The thesis advocates for further exploration of discrete diffusion models in combating online hate.

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ABBREVIATIONS

IITD	Indian Institute of Technology, Delhi
HS	Hatespeech
CS	Counterspeech
CONAN	Counter-Narrative
TST	Text Style Transfer
VQ-VAE	Vector Quantized Variational Autoencoder
SEDD	Score Entropy Discrete Diffusion Model
SEDD-BERT	Score Entropy Discrete Diffusion with BERT model
VQ-SEDD	Vector Quantized Score Entropy Discrete Diffusion Model
(P)	Pretraining
(IT)	Instruction Tuning

NOTATION

x_i	Hatespeech
y_i	Counterspeech
s_i	Style label
t_i	Target label
$\oplus$	Concatenation
sg	Stop Gradient
$z_e(x)$	Encoded Latent Vector
e_i	Codebook Vector
p_i	Personality encoded vector
d_i	Semantic encoded vector
C	Codebooks
μ_i	Style gate coefficient
β	Commitment loss coefficient

Chapter 1

INTRODUCTION

1.1 Motivation

The rise of digital communication has amplified the presence and impact of hate speech online, posing significant threats to individual well-being and the health of public discourse. Traditional moderation strategies, such as content removal or user bans, often struggle to balance the need for safety with the preservation of free expression, and may fail to address the root causes of harmful behavior [Ins22, ZRM23, JS25]. It is not feasible to control an entire population that enjoys constitutional freedom of speech through bans and deplatforming. In this context, **counterspeech** has emerged as a compelling alternative.

Counterspeech refers to direct responses to hateful or harmful speech, especially online, that aim to counteract, refute, or de-escalate such content through reasoned argument, empathy, humor, or factual correction. Counterspeech is needed because it can de-escalate conflicts, promote deradicalization, and encourage more peaceful, constructive interactions. Unlike punitive measures, it supports victims, signals community standards, and can influence not just the speaker but also the wider audience, gradually shifting norms and reducing the prevalence of hate [JS25].

Hatespeech: *Immigrants are ruining our country they should all be sent back!*

Counterspeech: *Immigrants have played a vital role in building our communities, contributing to healthcare, education and technology.Diversity can make us stronger, not weaker.*

The effectiveness of counterspeech hinges significantly on its perceived **credibility** and authenticity. Generic impersonal responses are often dismissed as automated or unintentional, failing to resonate with the target audience and thus limiting their persuasive impact [JSP23]. Research highlights that for counterspeech to be successful, it must be delivered in a voice that the audience perceives as trustworthy and authoritative, thereby increasing the likelihood of engagement and reflection [WG24].

However, most existing counterspeech, whether human or AI generated tends to be generic and impersonal, which can limit its effectiveness [SMBH24]. Research shows that **personalized counterspeech**, adapted to the specific context, user, and community, is more persuasive, adequate, and likely to foster genuine dialogue[CMT+25, MBL+24]. By tailoring responses to the individual and situation, counterspeech can build rapport, reduce defensiveness, and increase the likelihood of positive behavioral change.

Drawing inspiration from figures like **Mahatma Gandhi** and **Nelson Mandela** provides a powerful model for cultivating such a credible voice. These leaders are renowned for a communication style that is deeply personal, principled, and persuasive. Gandhi viewed speech as a tool to educate the public mind, rooted in the scientific precision of non-violence (*Ahimsa*) and truth-force (*Satyagraha*). His approach was not to inflict suffering on an opponent but to appeal to their conscience through his own deliberate and truthful words. He believed in an economy of words, ensuring his statements were carefully weighed and reflected a unique discipline. Similarly, Nelson Mandela, who was greatly influenced by Gandhian philosophy, championed non-violent resistance as an expression of moral strength and courage. His leadership emphasized dialogue, negotiation, and peaceful protest to bring about social change. Mandela also stressed the importance of empathy and reconciliation, believing that forgiveness was essential to heal a divided society. By adopting a persona inspired by these leaders, counterspeech can shift from an impersonal, **third-person rebuttal to a first-person** voice characterized by moral authority, empathy, and a commitment to peaceful dialogue.

Recent advances in discrete diffusion models have shown promising results in text generation tasks, rivaling the performance of autoregressive models such as GPT-2 [LME24, DG24]. Notably, the **Score Entropy Discrete Diffusion (SEDD)** model has demonstrated that diffusion-based approaches can achieve comparable or even superior perplexity scores while offering additional benefits such as iterative refinement, controllability, and improved sampling diversity. Unlike traditional autoregressive models that generate tokens sequentially and often suffer from exposure bias or unidirectional context limitations, discrete diffusion models operate by denoising tokens over multiple steps, enabling more holistic generation. The success of SEDD indicates that diffusion is not only viable for discrete text domains but may also provide a new paradigm for structured and controllable language generation[LME24].

In parallel, the increasing concern about **online hate speech** has required the development of intelligent and context-aware **counterspeech systems** that can defuse hostility without infringing on free speech. Traditional counterspeech methods often generate generic or templated responses, lacking personalization and thus failing to resonate with the target audience. This thesis explores the synergy between **diffusion-based text generation** and **personalized counterspeech**, aiming to create responses that are contextually nuanced and aligned with the persona or style of the responder. Using the structured denoising capabilities of diffusion models, we can better control aspects such as tone, user alignment, and semantic intent, making them a compelling choice for this socially impactful application.

1.2 Problem Statement

The rise of online hate speech presents a significant societal challenge, demanding responses that are not only assertive but also empathetic and unifying. Counterspeech is a non censoring response aimed at challenging hate and offers a powerful alternative to content removal. However, most existing counterspeech systems lack the persuasive depth and moral authority needed to effect meaningful change[ZB21, SMBH24]. This thesis focuses on generating effective counterspeech **in the rhetorical and philosophical style of leaders such as Mahatma Gandhi and Nelson Mandela**, who were known for their ability to unify, persuade, and de-escalate conflict through compassionate dialogue[SAnd, Wri20] .The goal is to produce responses that mirror their language style and values like nonviolence, tolerance, and justice, thus making counterspeech both impactful and emotionally resonant.

The problem is formalized as: given a **hate speech input** and a **desired persona** or style profile (e.g., Gandhi or Mandela), generate a **personalized counterspeech** response using a **diffusion-based language model**. The system must produce fluent, contextually appropriate, and ethically grounded responses that reflect the targeted persona's stylistic and moral tone. Key challenges include ensuring accurate **personalization** without drifting semantically, maintaining **grammatical and stylistic fluency** through multiple denoising steps, **conditioning discrete diffusion models** on rich text prompts, and enforcing **ethical constraints** to avoid reinforcing harmful narratives. Addressing these challenges is critical for building responsible and socially beneficial generative systems.

1.3 Research Objectives

The main objectives of this thesis are:

1. To collect a dataset of counterspeech examples in the rhetorical and stylistic tradition of Mahatma Gandhi and Nelson Mandela.
2. To develop a novel framework for personalised counterspeech generation using diffusion-based language modeling.
3. To explore the efficacy of pretrained encoder-only models like BERT with diffusion language modeling, to incorporate the pretrained knowledge.
4. To design evaluation metrics assessing the style control, response diversity, and persuasive effectiveness.
5. To compare baseline models empirically using both traditional metrics and human evaluation for qualitative assessment.

1.4 Contributions

The main contributions of this thesis are:

1. Collection and generation of data set of 10086 stylistic counterspeeches consisting of 5042 and 5044 responses in the style of Gandhi and Mandela, respectively.

2. We systematically investigate the application of diffusion models (SEDD, VQ-SEDD, and SEDD-BERT) for generating high-quality and contextually appropriate counter-speech.

3. Our experiments show that instruction-tuned diffusion variants outperform their non-instruction-tuned counterparts across numerous content, style, and novelty metrics, demonstrating the power of adding instructional signals to guide diffusion.

4. We propose VQ-SEDD, a two-phase approach where vector quantization is incorporated into diffusion, yielding a richer codebook representation. This results in more coherent, stylistically appropriate, and effective responses.

5. Our results show that diffusion models trained without large external pretraining can match or outperform many large, fine-tuned, or few-shot models. This highlights their ability to learn directly from task-specific data.

6. We present a multi-faceted evaluation, measuring content (BLEU, METEOR, BERTScore), style, diversity, novelty, toxicity, and quote match to thoroughly assess the capabilities of diffusion methods in this context.

7. Our approach consistently produces low-toxic, novel, and stylistically rich responses, outperforming many baseline models.

1.5 Thesis Organization

This thesis is organized into six chapters, which systematically present the research from its conception to its conclusion. Chapter 1 provides an introduction to the rising concern of hatespeech, establishing the context, motivation and significance of personalised counter-speeches in the style of Gandhi and Mandela. It outlines the primary research questions, defines the objectives, and summarizes the key contributions of this work, concluding with a roadmap for the subsequent chapters. Chapter 2 presents a comprehensive literature survey, critically reviewing existing work in counterspeech domain, Diffusion language modeling, Text style transfer(TST) and highlights significant research gaps. It situates our proposed approach within the broader academic landscape.

The core of our contribution is detailed in Chapter 3, which elaborates on the proposed Vector Quantized SEDD model(VQ-SEDD). This chapter provides an in-depth description of the VQ-SEDD model including the working of SEDD and VQ-VAE to address the research problem. Following this, Chapter 4 describes the Experimental Setup, detailing the datasets, training details, evaluation metrics, and implementation specifics used to validate our methodology. Chapter 5 presents the Results and Evaluation, where we report the empirical findings from our experiments. This chapter includes a thorough quantitative and qualitative analysis, comparing the performance of our method against established baselines and discussing the significance of the outcomes. Finally, Chapter 6 concludes the thesis by summarizing the key findings, revisiting the research objectives to demonstrate their fulfillment, acknowledging the limitations of the study, and suggesting promising directions for future work.

Chapter 2

LITERATURE SURVEY

2.1 Counterspeech

Counterspeech refers to direct responses to hateful or harmful speech, especially online, that aim to counteract, refute, or de-escalate such content without resorting to censorship or content removal. The goal of counterspeech is to address hate or misinformation by providing correct information, alternative perspectives, or appeals to empathy and reason, thereby preserving freedom of expression while combating harmful narratives. Counterspeech is often crowd-sourced, with ordinary users or trained experts responding to hate speech in public forums, social media, or comment sections.

Traditional methods to combat hate speech such as content deletion or user suspension, face criticism for potential overreach, censorship, and limited effectiveness. Counterspeech offers a non-censorial alternative that can be more flexible, responsive, and respectful of open discourse [BCAG24, MBL$^+$24]. International organizations and NGOs have promoted counterspeech as a preferred strategy, with platforms like Facebook and initiatives such as the Council of Europe's "No Hate Speech Movement" actively supporting counterspeech campaigns[Cou].

Counterspeech can take many forms, including factual rebuttals, highlighting hypocrisy, empathetic appeals, warnings of consequences, humor, and support for victims[JS25]. Research demonstrates that counterspeech not only supports victims but also influences bystanders and can reduce the likelihood of repeated hate speech by transgressors over time. Importantly, counterspeech aligns with international human rights principles and is considered more democratic and less prone to misuse than outright censorship[Cyp24].

Despite its promise, counterspeech faces practical barriers: individuals may lack the resources, training, or confidence to respond and may fear personal harm or doubt the impact of their interventions[MBL$^+$24]. These challenges underscore the need for scalable, effective and contextually appropriate counterspeech solutions, motivating research into automated and AI-assisted approaches. For example, an informative response may be best for correcting factual misinformation, while a denouncing tone can signal that a community norm has been violated. In other cases, using humor, empathy, or asking a question can be more effective at de-escalating a situation or encouraging reflection. Research demonstrates that tailoring the intent to the specific circumstances makes counterspeech significantly more adequate,

Strategy	Example
Facts	Actually, studies show that on the whole migrants contribute more to public finances than they take out, see this article for example.
Hypocrisy	Immigrants stealing British resources? A bit rich given how much was stolen from colonies by the British Empire.
Consequences	Spreading hateful content is illegal. Police will knock on your door.
Affiliation	As a British national, I know life is hard here right now. But I assure you that your unemployment is not the fault of immigrants.
Denouncing	Stop with the racist and derogatory slurs. It's unacceptable to talk this way.
Counter questions	Do you have a problem with all immigrants or only ones from lower income countries? Are you suggesting we have enough qualified and willing British born workers to fill all the jobs?
Humour	You should think about how the Spanish feel next time you go on holiday to Costa Del Sol (laughing emoji).
Positive tone	Immigrants strengthen UK society in so many ways - greater diversity, skillsets and innovation to name a few! And no way our NHS could function without the immigrant workforce.

Table 2.1: Types of Counterspeech Strategies for addressing the hate speech: "Immigrants are invading and stealing our resources"

relevant, and persuasive than generic replies, thereby increasing its effectiveness in mitigating online harm[GDG+23].

The complex and multifaceted nature of hate speech necessitates the use of different counterspeech intents, as a single, one-size-fits-all approach is often ineffective[CMT+25, GDG+23].For example, an informative response may be best for correcting factual misinformation, while a denouncing tone can signal that a community norm has been violated. In other cases, using humor, empathy, or asking a question can be more effective at de-escalating a situation or encouraging reflection [GDG+23, HKS+24]. Research demonstrates that tailoring the intent to the specific circumstances makes counterspeech significantly more adequate, relevant, and persuasive than generic replies, thereby increasing its effectiveness in mitigating online harm.

2.2 Diffusion Language Model

2.2.1 Continuous Diffusion for Text

Continuous diffusion models for text, first introduced by Li et al. (2022)[LTG+22], operate by embedding discrete tokens into a continuous latent space, learning a diffusion process within this space, and then mapping the generated outputs back to tokens via nearest neighbor search [OJ24, LTG+22]. Early iterations of these models faced challenges, but subsequent empirical advancements have led to notable improvements [GLF+23, TWZ+23].

Despite these innovations, continuous diffusion models face fundamental challenges when applied to text. The core issue is the mismatch between the continuous nature of the diffusion process and the inherently discrete structure of language [OJ24]. When mapping discrete tokens to a continuous space, informative signals can be lost during the transition, leading to degraded performance compared to discrete diffusion models. Additionally, continuous diffusion models for text generation often require many iterative denoising steps to approximate the reverse process, making them computationally expensive and slow during inference [GLF+23]. There are also notable gaps between training and inference: during inference, the model lacks access to the forward noise process, leading to exposure bias and error accumulation as denoising steps increase [TWZ+23]. These limitations highlight why continuous diffusion models, while theoretically appealing and effective for continuous data, struggle to match the efficiency and quality of discrete approaches in text generation tasks [GLF+23].

2.2.2 Discrete Diffusion Models for Text

Discrete diffusion models have emerged as a promising alternative to traditional autoregressive models for generating discrete data, such as natural language. While diffusion models have shown remarkable success in continuous domains, their extension to discrete spaces has been challenging due to the lack of suitable training objectives. The Score Entropy Discrete Diffusion model (SEDD) addresses this gap by introducing the score entropy loss, which generalizes score matching to discrete spaces [LME24]. Unlike standard score matching, which relies on Fisher divergences, score entropy is based on the Bregman divergence with a convex function. This formulation allows for the learning of ratios of data distributions, parameterizing the reverse diffusion process in discrete domains, and enabling effective training of discrete diffusion models.

Empirical results demonstrate that SEDD significantly outperforms previous language diffusion paradigms, reducing perplexity by 25-75% and even surpassing GPT-2 in certain benchmarks. Notably, SEDD generates high-quality, faithful text without requiring techniques like temperature scaling, achieving 6-8 times better generative perplexity than un-annealed GPT-2 [LME24]. The model also offers flexibility in trading off compute and quality, matching the quality of nucleus sampling with far fewer network evaluations, and supports controllable infilling strategies beyond left-to-right prompting. These advances position score entropy discrete diffusion models as a competitive and versatile approach for discrete generative modeling, particularly in language tasks.

2.3 Text Style Transfer

Text Style Transfer (TST) has emerged as a significant and challenging subfield of controlled text generation. The primary goal of TST is to modify the stylistic attributes of a given text, such as its politeness, formality, or sentiment while preserving its core semantic content. This task is distinct from more general personalized generation, as it typically **operates on a predefined and explicit set of style constraints**. Much of the research in this area has focused on transferring discrete attributes, for instance, converting a negative movie review into a positive one or rephrasing an informal sentence into a formal equivalent. The methodologies for TST are often categorized based on the type of training data available: those that rely on parallel corpora (paired sentences with the same content but different styles) and those designed for non-parallel settings, which are more common but present a greater technical challenge.

The foundations of TST are rooted in broader techniques for controlled text generation, which aim to steer the output of large language models (LMs). Researchers have developed several innovative approaches to achieve this control. A novel approach by Dathathri et al. (2020) introduced Plug and Play Language Models (PPLM), which guide a pre-trained model like GPT-2 by manipulating its internal latent representations at inference time, thereby avoiding the need for costly fine-tuning [DMC+20]. Another powerful, parameter-efficient technique is Prefix-Tuning, proposed by Li & Liang, which freezes the main LM and instead learns a small, task-specific prefix vector that is prepended to the input sequence to steer the model's generation process [LL21]. Earlier work also demonstrated control at the decoding stage; for example, the Hafez system by Ghazvininejad et al. (2017) generated poetry in a desired style by directly adjusting the sampling weights during beam search [GSCK17]. These methods provide the essential mechanisms that can be adapted and refined for the specific demands of text style transfer.

2.4 Research Gaps

Need for Personalized Counterspeech in the Style of Gandhi and Mandela

Personalized counterspeech inspired by the empathetic, nonviolent communication styles of Mahatma Gandhi and Nelson Mandela is a crucial yet underexplored area in combating hate speech. While counterspeech is widely recognized as an effective and democratic alternative to censorship [Ben19, MFF21], current approaches often lack the credibility and emotional appeal necessary to foster genuine dialogue and long-term attitude change [WH22]. Gandhi and Mandela exemplified how addressing adversaries with dignity, appealing to shared humanity, and employing nonviolent resistance can diffuse hostility and inspire societal transformation [Man94, Par97]. However, existing automated or crowd-sourced counterspeech solutions rarely incorporate such nuanced, individualized strategies, limiting their impact [Ben19, WH22]. The absence of AI systems capable of emulating these iconic leaders' values, tone, and persuasive techniques represents a significant research opportunity, especially given the cross-cultural resonance and effectiveness of their approaches in conflict resolution [Man94, Par97].

Limitations of Autoregressive Language Models for Diverse and Controllable Text Generation

A prominent research gap lies in the limitations of standard autoregressive language models, such as GPT-2 and its variants, when tasked with generating diverse, contextually controlled, and high-fidelity text for sensitive applications like counterspeech [DGSS21, ZSL+23]. Autoregressive models are prone to issues such as exposure bias and error accumulation, where mistakes made early in the generation process cascade and degrade the quality of subsequent output [HZZG21, Wen21]. Because they generate text sequentially, they often struggle to maintain global consistency and capture complex, long-range dependencies, which can result in repetitive or generic text [Gao19, Wen21].

These are not engineering problems that can be solved simply by scaling up model size or training data; they are intrinsic architectural shortcomings, as autoregressive modeling can be a misspecified objective for text [DGSS21]. For applications requiring nuanced style transfer, such as emulating the rhetorical approaches of Gandhi or Mandela, these models lack explicit mechanisms for control over tone, intent, or persuasive strategy, making them ill-suited for generating personalized, impactful counterspeech [MLL+24, ZSL+23, WZZ+24]. This underscores the need for alternative generative frameworks such as diffusion-based or latent-variable models that can offer greater flexibility, diversity, and controllability [Pag25, She21], and can be tailored to ethically and effectively address hate speech in a manner that honors the legacies of transformational leaders.

Need for Metrics Evaluating Relevance and Stylistic Alignment of Counterspeech

A further research gap exists in the development of robust, multi-dimensional metrics for evaluating counterspeech, particularly with respect to its relevance to the target and alignment with specific persuasive styles, such as those of Gandhi or Mandela. Current evaluation practices often rely on standard NLP metrics like BLEU, ROUGE, or BERTScore, which primarily assess surface-level similarity and fail to capture deeper aspects such as contextual relevance, argumentative coherence, or stylistic fidelity [HKBC25, CYC23]. Human evaluations, while valuable, are labor-intensive and lack scalability [HKBC25].

Recent frameworks, such as CSEval, have begun to address these limitations by introducing dimensions like contextual relevance, coherence, aggressiveness, and suitableness [HKBC25]. However, there remains a lack of automated, reference-free metrics that can reliably assess whether counterspeech is not only contextually appropriate but also delivered in a manner consistent with a desired rhetorical style [HX25, MKMB22]. This gap is especially pronounced for personalized counterspeech generation, where measuring both the effectiveness and the stylistic authenticity of responses is crucial for real-world impact and user trust [MKMB22].

Chapter 3

Methodology

3.1 Score Entropy Discrete Diffusion

It is a discrete diffusion language modeling approach, which uses **mask denoising objective** instead of next-word prediction for training the transformer model. *The mask denoising objective teaches the language model to unmask or predict original tokens in place of masked ones.* It encourages the model to use the left as well as right context of non-corrupted tokens. Developed by Lou, Meng, and Ermon[LME24], SEDD introduces score entropy, a novel training loss that generalizes continuous score matching to discrete spaces, enabling high-quality text generation with unique algorithmic advantages.

In the continuous domain, score matching involves estimating the score, which is the gradient of the log probability density with respect to the data. This guides the parameters to move in the direction of higher probability density.The analogous concept in the discrete domain can be represented by the ratio of forward and backward transition probabilities, reflecting how likely it is for a state to move to another state and back. Instead of computing a literal gradient, this ratio expresses a form of score for the likelihood of transitions between states.

3.1.1 The Forward Diffusion Process

The model first defines a forward diffusion process as continuous time markov process that evolves a probability distribution over a finite set of states, $X = \{1, \ldots, N\}$. This process transforms an initial data distribution, p_0, which approximates the true data distribution (p_{data}), into a simple base distribution, p_{base}, as time $t \to \infty$.

This evolution is described by a linear ordinary differential equation (ODE):

$$\frac{dp_t}{dt} = Q_t p_t \tag{3.1}$$

In this equation, the components are:

- $p_t \in \mathbb{R}^N$: A probability mass vector at time t. Its entries are positive and sum to 1, representing the probability distribution over the states in X.
- $Q_t \in \mathbb{R}^{N \times N}$: The diffusion matrix (or rate matrix) at time t. This matrix has two critical properties:

1. Its non-diagonal entries are non-negative ($Q_t(x, y) \geq 0$ for $x \neq y$).
2. Its columns sum to zero ($\sum_{x=1}^{N} Q_t(x, y) = 0$ for all y). This ensures that the total probability mass is conserved over time, so p_t remains a valid probability distribution.

On solving ODE given in Equation 3.1 :

$$\frac{dp_t}{dt} = Q_t p_t$$

with initial condition:

$$p_0 = x_0$$

As we know it is a linear homomgenous ODE, and its solution is given by:

$$p_t = e^{Qt} x_0$$

Here e^{Qt} denotes the matrix exponential, defined by the power series expansion:

$$e^{Qt} = I + Qt + \frac{1}{2!}Q^2t^2 + \frac{1}{3!}Q^3t^3 + \cdots ,$$

One can simulate the forward process by taking small Δt euler steps and randomly sampling the resulting transitions. In particular, the samples are defined by transition densities which come from the columns of Q_t:

$$p(x_{t+\Delta t} = y \mid x_t = x) = \delta_{xy} + Q_t(y, x)\Delta t + O(\Delta t^2),$$

where

$$\delta_{xy} = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases},$$

3.1.2 The Reverse Generative Process

The power of a diffusion model lies in its ability to reverse this process. Starting from a sample of the simple base distribution p_{base} at a large time T, the model can generate a sample from the complex data distribution p_0 by evolving backwards in time.

This reverse process is also a continuous-time Markov process, governed by a related time-dependent ODE [SYD+23]:

$$\frac{dp_{T-t}}{dt} = \bar{Q}_{T-t} p_{T-t} \tag{3.2}$$

with initial condition:

$$p_T = x_T \approx p_{base}$$

Here, $\bar{Q}_t$ is the reverse diffusion matrix. Its components are defined in terms of the forward matrix Q_t and the time-dependent probability distributions p_t. The off-diagonal entries are given by Feller's theorem for the time-reversal of a Markov process:

$$\bar{Q}_t(y, x) = \frac{p_t(y)}{p_t(x)} Q_t(x, y) \quad (\text{for } y \neq x) \tag{3.3}$$

The diagonal entries, $\bar{Q}_t(x, x)$, are then defined to ensure the columns of the reverse matrix also sum to zero:

$$\bar{Q}_t(x, x) = -\sum_{y \neq x} \bar{Q}_t(y, x) \tag{3.4}$$

For our time-dependent score network $s_\theta(\cdot, t)$, the parameterized reverse matrix is defined as:

$$Q_\theta^t(y, x) = s_\theta(x, t)_y Q_t(x, y),$$

for $x \neq y$,

Reverse Diffusion process can be simulated using Euler strategy as,

$$p(x_{t-\Delta t} = y \mid x_t = x) = \delta_{xy} + s_\theta(x, t) Q_t(x, y) \Delta t + O(\Delta t^2),$$

where

$$\delta_{xy} = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases},$$

3.1.3 Score Entropy Loss

At its heart, SEDD learns the **ratios of the data distribution** $\frac{p_t(y)}{p_t(x)}$ through a loss function called **score entropy**. This approach directly models transition probabilities between discrete states (e.g., tokens). The score entropy loss is defined as:

$$\mathcal{L}_{\text{SE}} = \mathbb{E}_{x \sim p} \left[\sum_{y \neq x} w_{xy} \left(s_\theta(x)_y - \frac{p(y|x_0)}{p(x|x_0)} \log s_\theta(x)_y + K \left(\frac{p(y|x_0)}{p(x|x_0)} \right) \right) \right] \tag{3.5}$$

where

$$K(a) = a(\log a - 1) \tag{3.6}$$

The importance of score entropy loss function is that it minimizes at score value as ratio of probability of states, which can be used to learn the parameters Θ of a transfomer based model through backpropagation.

Algorithm 1 Score Entropy Training Loop

Require: Network s_θ, noise schedule σ (total noise $\overline{\sigma}$), data distribution p_{data}, token transition matrix Q, time $[0, T]$.

Sample $x_0 \sim p_0$, $t \sim \mathcal{U}([0, T])$.

Construct x_t from x_0. In particular, $x_t^i \sim p_t(\cdot|x_0^i) = \exp(\overline{\sigma}(t)Q)x_0^i$.

if Q is Absorb **then**

  This is $e^{-\overline{\sigma}(t)} e_{x_0^i} + (1 - e^{-\overline{\sigma}(t)}) e_{\text{MASK}}$

else if Q is Uniform **then**

  This is $\frac{e^{\overline{\sigma}(t)} - 1}{n e^{\overline{\sigma}(t)}} \mathbb{1} + e^{-\overline{\sigma}(t)} e_{x_0^i}$

end if

Compute

$$\hat{\mathcal{L}}_{DW\,DSE} = \sigma(t) \sum_{i=1}^{d} \sum_{y=1}^{n} (1 - \delta_{x_t^i}(y)) \left(s_\theta(x_t, t)_{i,y} - \frac{p_{t|0}(y|x_0^i)}{p_{t|0}(x_t^i|x_0^i)} \log s_\theta(x_t, t)_{i,y} \right).$$

Backpropagate $\nabla_\theta \hat{\mathcal{L}}_{DW\,DSE}$. Run optimizer.

This algorithm outlines a training procedure for a neural network s_θ using score entropy, designed for discrete-state diffusion or token transition models.

1. **Input Setup:** The network s_θ learns to model token transitions governed by a noise schedule $\sigma(t)$ and transition matrix Q. Data samples x_0 are drawn from the dataset p_{data}, and timesteps t are uniformly sampled from $[0, T]$.

2. **Noise Corruption:** At each timestep t, input tokens x_t are corrupted using either:
   - Absorbing transitions: Tokens gradually mask out with probability $1 - e^{\sigma(t)}$.
   - Uniform transitions: Tokens blend with a uniform distribution over n classes, controlled by $e^{\sigma(t)}$.

3. **Loss Calculation:** The custom loss L_{DWDSE} penalizes the network's predictions $s_\theta(x_t, t)$ by comparing them to the analytical ratio $\frac{p_{t|0}(y|x_0^i)}{p_{t|0}(x_t^i|x_0^i)}$, excluding the already corrupted token x_t^i. The $\sigma(t)$ term scales the loss according to the noise intensity.

4. **Optimization:** Gradients of this loss are backpropagated to update s_θ, training it to invert the corruption process by estimating score-like adjustments for non-corrupted tokens.

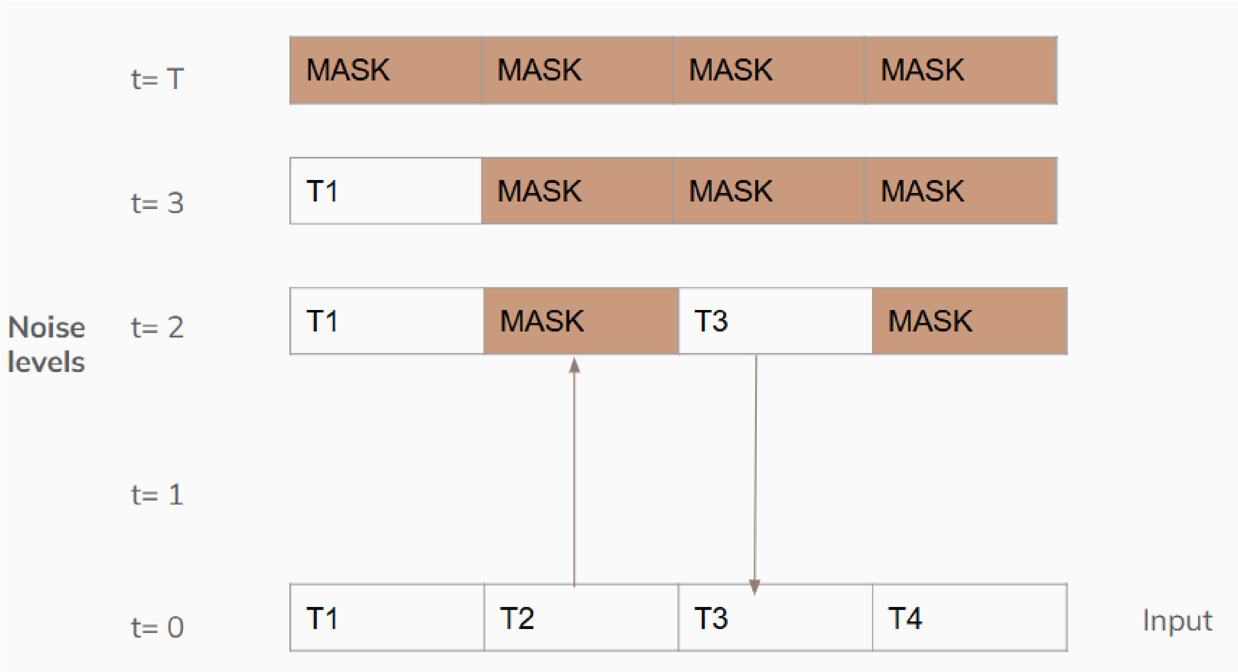

Figure 3.1: Discrete Diffusion: Single Training step

3.2 Vector Quantization

Vector quantization is the process of converting **continuous latent representations from the encoder into discrete codes** by mapping them to the nearest vector in a learned codebook of embeddings. It is widely used in Vector Quantized Variational Autoencoders(VQ-VAE) on image data producing **clustered features** in the form of **codebooks**. Many real-world data types (like language, phonemes in speech, or image patches) are inherently discrete. Vector quantization allows the model to represent data using a finite set of learned codebook vectors, making it possible to use these representations as categorical features for downstream models.

Key Components of VQ-VAE

- **Encoder**: Maps input x to a continuous latent vector $z_e(x)$.
- **Codebook**: A set of K learnable vectors $\{e_1, e_2, \ldots, e_K\}$.
- **Quantization**: Replace $z_e(x)$ with the nearest codebook vector:

$$z_q(x) = e_k \quad \text{where } k = \arg\min_j \|z_e(x) - e_j\|_2$$

- **Decoder**: Reconstructs input from $z_q(x)$.

Loss Function

The total loss combines three terms:

$$\mathcal{L} = \underbrace{\log p(x|z_q(x))}_{\text{Reconstruction Loss}} + \underbrace{\|\text{sg}(z_e(x)) - e_k\|_2^2}_{\text{Codebook Loss}} + \underbrace{\beta\|z_e(x) - \text{sg}(e_k)\|_2^2}_{\text{Commitment Loss}}$$

Reconstruction Loss

- **Purpose**: Ensures the decoder accurately reconstructs the input x from the quantized latent $z_q(x)$.
- **Details**:
  - Mathematically equivalent to the negative log-likelihood of the input given the quantized latent (e.g., MSE for images, cross-entropy for discrete data).
  - Trains the decoder and encoder via gradients propagated through the **straight-through estimator** (bypassing non-differentiable quantization).

Codebook Loss

- **Purpose**: Updates codebook vectors e_k to better match encoder outputs $z_e(x)$.
- **Details**:
  - The stop-gradient operator sg prevents gradients from flowing back to the encoder, ensuring only codebook vectors are updated.
  - Minimizes the distance between codebook entries and encoder outputs, effectively clustering encoder outputs around codebook vectors.

Commitment Loss

- **Purpose**: Encourages the encoder to produce outputs close to codebook vectors, preventing instability.
- **Details**:
  - β (typically 0.25) balances the influence of this term.
  - The stop-gradient on e_k ensures only the encoder is updated, anchoring its outputs near the codebook and avoiding drift.

We intend to utilise the vector quantization concept for capturing better style representations in the discrete diffusion model, which can help us generating better stylistic text.

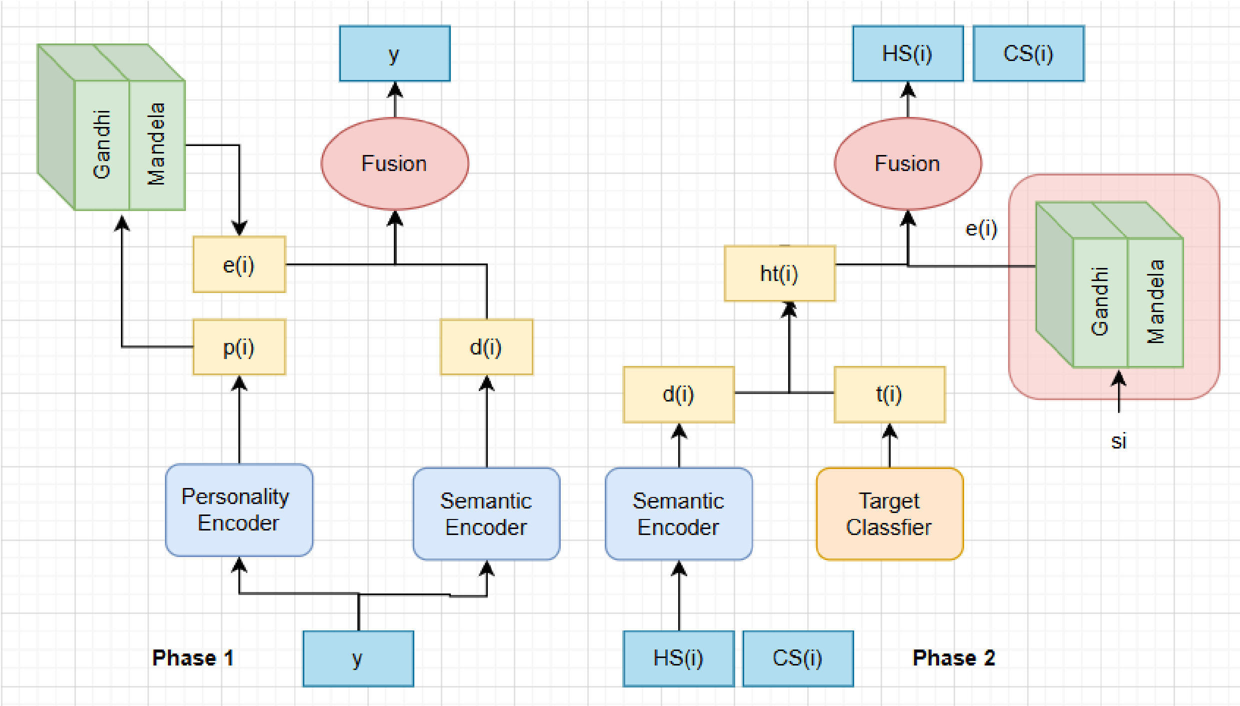

Figure 3.2: Architecture of the proposed VQ-SEDD model. It consist of two phases, codebook learning and counterspeech generation.

3.3 Proposed Approach

Let us denote the dataset $D = \{(x_1, t_1, s_1, y_1), ..., (x_n, t_n, s_n, y_n)\}$, where x_i denotes the i^{th} hate-speech instance, t_i denotes the target of x_i, y_i denotes the counterspeech corresponding to x_i and s_i denotes the style of y_i. We introduce the Two-Phase Vector-Quantized SEDD (VQ-SEDD) model for generating stylistic counterspeech.

3.3.1 Phase 1: Codebook Learning

This stage is responsible for **capturing the style representation** of the counterspeech using vector quantization with forced alignment. The style of the counterspeech includes the *structure like the acknowledgement in the starting and the quotes (inside double quotes) which are used for responding to the hatespeech.* We keep two Encoder-only models like BERT, personality (style) encoder and semantic encoder. We keep them separate to keep the style apart from the semantic content of the text. Refer to Fig3.2 .

Personality Encoder

The counterspeech y_i is first tokenized into sub-word embeddings $y_i^t \in R^{n \times D}$, where n is the maximum sequence length and D is the latent dimension of the model, which is 768 for BERT. The tokenized embedding y_i^t is passed through a personality encoder to obtain personality encoding p_i^t. To learn a quantized distribution of style, we used codebooks similar to VQ-VAE, but instead of reconstruction loss, we replaced it with score entropy loss. The style-codebooks $C \in R^{|C| \times d}$, is a matrix with each row correspond to one style representation. Here $|C| = 2$, representing counterspeeches in the style of Gandhi and Mandela with $d = 768$ compatible with the BERT encodings. The p_i encoding is replaced by the corresponding codebook e_i. The modified loss function uses **score entropy loss to reconstruct the text** and codebook loss, commitment loss moves the codebook closer to encoded latent vectors as well as encodings towards the codebook.

Forced Alignment: We fix the codebook indexes of leaders and always select the corresponding codebook instead of computing the distance between the hidden state and the codebooks, so as to capture the features of one leader in one codebook.

Semantic Encoder

The responsibility of this encoder is to preserve the semantic information in the text. The tokenized embedding y_i^t is passed through a SEDD-BERT encoder model to obtain a diffusion encoding d_i^t.

This is the combined loss function for phase 1,

$$\mathcal{L} = \mathcal{L}_{\text{SE}} + \|\text{sg}(p_i) - e_i\|_2^2 + \beta\|p_i - \text{sg}(e_i)\|_2^2 \tag{3.7}$$

Gated Fusion Mechanism

The learned codebook e_i^t is fused with the diffusion encoding d_i^t with an adaptive gating mechanism and then the masked tokens are unmasked using a projection layer over entire vocabulary.

Given, the diffusion encoding $d_i \in R^{B \times n \times d}$ and codebook $e_i \in R^{B \times d}$, where B is the batch size, n is the sequence length and d is the model dimension.

1. **Concatenation and Transformation**

$$\mathbf{C}^{(0)} = [d_i; e_i] \in \mathbb{R}^{B \times n \times 2d}$$

$$\mathbf{C}^{(l)} = \mathbf{C}^{(l-1)} + \text{Layer}^l(\mathbf{C}^{(l-1)}), \quad l \in \{1, \ldots, L\}$$

   where each $Layer^l$ contains Linear $\rightarrow$ GELU $\rightarrow$ $Layer_{Norm}$.

2. **Adaptive Gating**

$$\mu_i = \sigma(\mathbf{W}_i \mathbf{C}^{(L)}) \quad \text{(Style gate)}$$

where σ is sigmoid and W_i are learned weights.

3. **Gate-Controlled Fusion**

$$\mathbf{Z}_{\text{int}} = (1 - \mu_i) \odot d_i + \mu_i \odot e_i$$

4. **Final Composition**

$$\mathbf{Z}_{\text{fused}} = d_i \odot \mathbf{Z}_{\text{int}} + \mathbf{C}_{:d}^{(L)}$$

3.3.2 Codebook Learning Approach

We used two codebook learning methods for training with different tasks. Refer Table 3.1.

- **VQ-SEDD-1**
  - Learning to generate counterspeech from an empty string.
- **VQ-SEDD-2**
  - Learning to generate two identical counterspeech seperated by a delimiter.

Model	Type	Input	Output
VQ-SEDD-1	Training	CS	CS
	Inference		CS
VQ-SEDD-2	Training	CS[SEP][CS]	CS[SEP]CS
	Inference	CS[SEP]	CS[SEP][CS]

Table 3.1: Two codebook Learning approaches.

3.3.3 Phase 2: Counterspeech Generation

This stage is responsible for *learning the denoising such that hatespeech x_i is followed by counterspeech y_i in the particular style s_i and should be focused on the target t_i in the opposite stance.* We use the concatenated text, *combined* $= x_i \oplus s_i \oplus t_i \oplus y_i$, as input to the diffusion encoder trained in the first phase. Refer to 3.2

Additionally, we use a **target classifier** to guide the model in generating counterspeech on the particular target. We also add the learned **target embedding** t_i to the diffusion encoding d_i to form ht_i. The combined ht_i is fused with the codebook e_i using the fusion mechanism with weights being initialized from the first phase. We also reuse the codebooks learned in the first phase but freeze them in the second phase. We train the diffusion BERT with the adaptive gating fusion mechanism and jointly target classifier to learn the parameters for generating counterspeech.

This is the combined loss function for phase 2,

$$\mathcal{L} = \mathcal{L}_{\text{SE}} - logp(t_i|x_i) \tag{3.8}$$

3.3.4 Instruction Tuning

This instruction tuning approach involves fixing a certain number of tokens at the beginning of each input sequence, these are the context tokens. During both training and inference, these initial tokens remain unmasked and unmodified as shown in red color in 3.3, serving as the contextual anchor for the model. The remaining tokens in the sequence are masked, and the model is trained to predict (unmask) these tokens based on the fixed context.

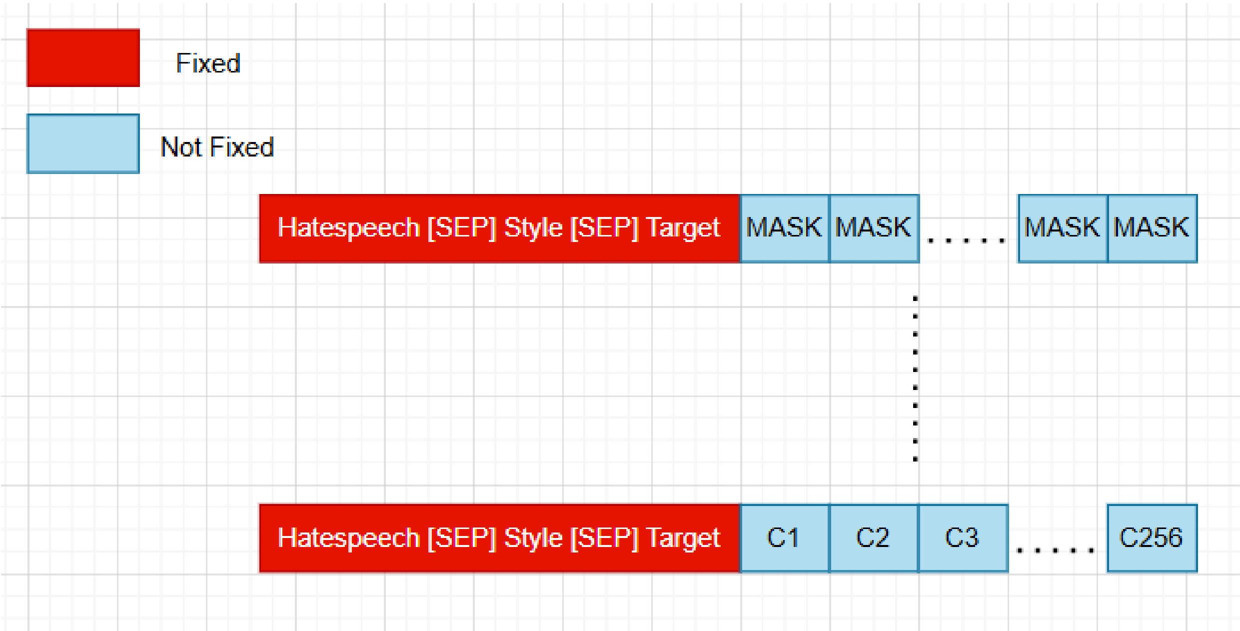

Figure 3.3: Procedure of Instruction tuning

Key Steps:

1. **Input Construction**

   Each training sample is divided into two parts: a fixed prefix of hatespeech, style, target and a sequence of masked tokens representing the completion or response to be generated.

2. **Masking Strategy**

   During training, only the completion tokens are masked and included in the loss calculation. The context tokens are excluded from the loss, ensuring the model focuses on generating the completion conditioned on the given hatespeech context and writing style.

3. **Inference**

   At inference time, the same number of initial tokens is provided as fixed context, and the model generates or unmasks the remaining tokens, using the context to ensure relevance and coherence in its output.

3.4 Designed Metrics

3.4.1 Style Score

This metric quantifies the stylistic alignment of a counterspeech with the writing of Gandhi versus Mandela, presented as a normalized score between 0 and 1. Moreover, we removed the quote from the counterspeech, to ensure the model does not have bias towards fixed number of quotes.

We trained a roberta-large model for 3 epochs using the AdamW optimizer with a learning rate of 5×10^{-5} and a linear learning rate scheduler. The model's performance is evaluated using 5-fold stratified cross-validation to avoid overfitting of the model. We achieve a 98% accuracy in accurately classifying the counterspeech style.

3.4.2 Quote Matching Score

It is a normalized Levenshtein score between 0 and 100, calculated between the generated quote in the counterspeech and the quote in the reference counterspeech.

We extract the quotes which are the text inside the double quotes using regular expression. The distance between the quotes is quantified using Fuzz score, which is a normalised Levenshtein distance.

$$\text{Fuzz Score} = \left(1 - \frac{\text{Levenshtein}(s1, s2)}{\max(|s1|, |s2|)}\right) \times 100$$

where s1 and s2 are the generated and reference counterspeech.

3.4.3 Effectiveness Score

This metric quantifies the effectiveness and relevance score of the generated counterspeech with respect to the particular hatespeech, style and target, presented as a normalized score between 0 and 1. The model receives input formatted as a sentence pair.

- Sentence 1: A concatenation of the hate speech's style, its target, and the hate speech content itself. Format: (Style[SEP]Target[SEP]HS)
- Sentence 2: The potential counterspeech. Format: CS

The model outputs a prediction for one of two labels:

- Label 1 (Effective): The `CS` is a relevant and appropriate response to the `HS`.

- Label 0 (Ineffective): The `CS` is an irrelevant or randomly chosen response to the `HS`

Positive Pairs (Label 1):

- Directly uses authentic HS-CS pairs from the source data
- Preserves original relationships between:
  - Hate speech content and counterspeech response
  - Target group alignment (`Target` column)
  - Communication style consistency (`Style` column)

Negative pairs (Label 0)

- Random HS-CS pairs
- Style Mismatch
- Target Mismatch

The RoBERTa-large model was fine-tuned for 6 epochs with an AdamW optimizer (lr=2×10^{-6}) and linear learning rate scheduling. To address the 2:5 class imbalance between effective and ineffective counterspeech pairs, class weighting (0.7:0.3) was applied to the cross-entropy loss function, prioritising recall of the underrepresented positive class.

A 5-fold stratified cross-validation approach ensured balanced style and target distributions across training/validation splits and strict model evaluation. This configuration achieved 90% F1-score in validation, deliberately chosen over accuracy as the primary metric due to its robustness in imbalanced classification scenarios. The evaluation emphasized precise identification of contextually relevant counterspeech while maintaining style and target alignment.

Chapter 4

Experimental Setup

4.1 Dataset

The dataset consists of 10086 rows with 5042 and 5044 style labels of Gandhi and Mandela respectively. The hatespeeches used for counterspeech generation are collected from Intent-CONAN and Multi-Target CONAN datasets[GDG+23, Fan21]. Initially, counterspeeches have been generated using Google Gemini API and then annotated for appropriateness, stylistic alignment and quote matching by 5 annotators. It is designed to facilitate the study and development of automated systems that generate nonviolent, truth-centered, and dignified counterspeech to hate speech, reflecting the leader's philosophical and rhetorical style of persuasion.

4.1.1 Dataset Description

The dataset generated consists of structured counterspeech responses to hate speech inputs, crafted in the distinct rhetorical and philosophical styles of Mohandas K. Gandhi and Nelson Mandela. Each data entry is a single row in a CSV file, with the following columns:

- **id**: A unique identifier for each sample.
- **hatespeech**: The original hate speech statement.
- **relevant_quotes**: A list of quotes from either Gandhi or Mandela, relevant to the hate speech topic.
- **core_issue**: A concise identification of the underlying emotional or ideological root of the hate speech (e.g., fear, resentment, misinformation, prejudice).
- **contextual_reframing**: A reinterpretation of the issue from the perspective of Gandhi or Mandela, emphasizing their core principles.
- **quote_justification**: An explanation for the selection of a particular quote, showing how it addresses the core issue.
- **style**: Label for the counterspeech style (Gandhi/Mandela)
- **target**: Label for the target group (9 classes)
- **counterspeech**: A 150–250 word response in the authentic voice of Gandhi or Mandela, following strict stylistic and ethical guidelines, addressing the hate speech and offering a constructive, dignified alternative.

The prompts for generating Gandhi-style and Mandela-style counterspeeches are carefully designed to reflect the unique rhetorical and philosophical traditions of each leader. Gandhi-style prompts emphasize **satya (truth), ahimsa (nonviolence), humility, and spiritual wisdom**, using simple, gentle language, metaphors from daily life, and references to **Indian philosophy**. They focus on **universal ethics, shared humanity, and self-examination**, with a gentle correction of misinformation and an appeal to conscience and hope. In contrast, Mandela-style prompts are grounded in **justice, reconciliation, dignity, and pragmatic moral courage**, employing clear, direct, and dignified language that draws on **South African experience** and the concept of **ubuntu**. Mandela's responses recognize pain and anger without humiliation, and stress themes of forgiveness, transformation, unity, and resilience. They correct misinformation assertively, appeal to **collective action**, and always justify the use of Mandela's quotes based on the core issue. These differences ensure that the counterspeech dataset authentically captures the distinct voices and strategies of Gandhi and Mandela.

To evaluate the generated responses, the dataset was annotated by five independent expert annotators with background in social science, online hate speech and published papers in related domain or completed advanced studies. The evaluation focused on multiple dimensions: the appropriateness of the counterspeech in response to the hateful message, the stylistic alignment with the chosen persona (i.e., how closely it resembled the communication style of Gandhi or Mandela), and the factual and relevance accuracy of any included quotes. All 3 dimensions were rated in the range of 1 to 5.

To measure the reliability of our human judgments, we calculated the inter-annotator agreement (IAA). We attained an average pairwise cohen's **Kappa score** of around **0.83** in all 3 metrics, indicating substantial agreement among the annotators. Refer to Appendix B.0.1 for pairwise Cohen's Kappa Score between the 5 annotators. Any initial disagreements in the ratings were resolved through discussion to reach a final consensus. This rigorous verification process was particularly crucial for the stylistic alignment, which was comprehensively studied through articles, letters and books for understanding the philosophy and rhetorical style of arguments of the two leaders. Two sessions of annotations are discussed in detail in Appendix B.0.1.

Example -

Hatespeech: Women should not to be allowed to vote.

Counterspeech Example(Gandhi): My dear friend, I hear your words, but I must gently disagree. To suggest that women should not vote is to misunderstand the very essence of equality and justice. "**To call women the weaker sex is a libel; it is man's injustice to women.**" They are not weaker, but possess strength and wisdom that is vital to our society. The vote is a sacred right, a voice that shapes our shared future. *How can we claim* to be a truly self-governing nation if we silence half of our population? *Imagine a family where only the father speaks is that a balanced and harmonious home?* No. It is the same with our nation. Let us embrace the strength and wisdom of women, and together, build a more just and peaceful world. **Reflect**, my friend, on the inherent dignity of every human being, regardless of gender.

Counterspeech Example(Mandela): My compatriot, I hear your words, but I must respectfully disagree. The exclusion of women from the right to vote is a grave injustice, a denial of their inherent dignity and their essential role in building a just society. As I have said, "**The fight against apartheid liberated all South Africans - Africans, coloureds, Indians, and whites - the struggle for gender equality will benefit both men and women. The prosperous future to which we aspire calls for a united front of all South Africans across both the colour and gender divides.**" To deny women their voice is to cripple our nation, to silence half of our people. Our journey towards freedom demands that we embrace equality in all its forms. Let us, therefore, **work together** to ensure that every citizen, regardless of gender, has the right to shape our collective destiny. Only then can we truly call ourselves a free and democratic society.

4.1.2 Model Architecture

All the SEDD model variants experimented in this paper uses the **pretrained BERT model** architecture with 110 M parameters. The BERT configuration is slightly modified to accommodate an additional **absorbing state** in the vocabulary, which is relevant for the diffusion process[LME24].

The style conditioning is achieved through a vector quantization module. This module learns **2 discrete codebooks** of style embeddings. During training, the model can be forced to use ground-truth style labels, while at inference, it selects the closest style embedding via nearest neighbor search. The quantizer also computes a commitment loss to ensure stable codebook usage. VQ-SEDD model uses score entropy loss and vector quantization loss while learning codebooks and only score entropy loss along side target classifier loss while generting counterspeech in second phase.

During both training phases, the model employs a gated fusion mechanism controlled by a style gate. The fused output after this mechanism is passed through a projection layer, which maps the features to the BERT vocabulary size of 30,522.

To model the progression of the diffusion process, a TimestepEmbedder generates sinusoidal embeddings of the diffusion timestep (sigma)[LME24]. These embeddings are added to the BERT outputs, allowing the model to condition its predictions on the current diffusion stage

4.1.3 Training Details

The dataset used in these experiments is divided into training, validation and test subsets in a 72 %,8 % and 20 % ratio respectively, and the split is made reproducible by setting the random seed to 42.

All SEDD model variants are trained from scratch for 50 epochs using a batch size of 32, a learning rate of 0.001, and a weight decay of 0.001, leveraging a single Nvidia A100 GPU for computation. During training, a log-linear noise schedule is employed, with 128 steps of noising and denoising, following the methodology outlined in the original SEDD paper.

For baseline models evaluated in **zero-shot and few-shot settings**, text generation is performed with a maximum of 256 new tokens, which aligns with the maximum sequence length used for counterspeech generation. Additionally, sampling is conducted deterministically (with do_sample set to False) to ensure a fair comparison with the diffusion-based models, as this disables stochastic sampling and annealing. These baseline models, having already been pretrained on large-scale corpora, are further pretrained for 4-5 epochs on the task-specific data considering the size of training set, and supervised fine-tuning for 3 to 5 epochs with context length of 256 and a learning rate of 2×10^{-5} and a weight decay of 0.03. An exception is made for the LLama 3.2 1B models, which are fine-tuned for 12 epochs under otherwise similar conditions.

4.1.4 Evaluation Metrics

The following metrics have been used for evaluation-

- **ROUGE-1:** Measures the overlap of unigrams (single words) between the generated text and the reference text, quantifying word-level similarity.
- **ROUGE-2:**Calculates the overlap of bigrams (pairs of consecutive words) between the candidate and reference, reflecting short phrase similarity.
- **ROUGE-L:**Evaluates the longest common subsequence between the generated and reference texts, assessing the preservation of in-order word sequences.
- **METEOR:**Computes a harmonic mean of unigram precision and recall, incorporating stem, synonym, and paraphrase matches for nuanced evaluation.
- **BERTScore:** Uses contextual embeddings from BERT to measure the semantic similarity between candidate and reference texts via cosine similarity of aligned word vectors.
- **STYLEScore:** Uses the trained roberta-large model for evaluating the stylistic alignment of a counterspeech.
- **Effectiveness:** Uses the trained roberta-large model for evaluating the effective relevance of the counterspeech.
- **Quote Match:** Uses Fuzz score to compute the normalised edit distance between the generated and reference quote.
- **Novelty:** Quantifies the proportion of words in the hypothesis that do not appear in the reference, penalized for low diversity, short length, and lack of semantic overlap.
- **Diversity:** Measures word variety in text using unique-to-total word ratio with penalties for repetition, short length, and incoherence, returning a normalized 0-1 score.

Chapter 5

Results and Evaluation

5.1 Experiments

- Experiment 1: (Pretrained Model Architectures)
  - Exploring the effect of pretrained model like BERT instead of existing GPT2 like transformer architecture of SEDD model.
  - **Intuition**: Pretrained knowledge can provide more context to the model.
  - **Result**: SEDD-BERT outperforms SEDD across all metric indicating the correlation of pretrained knowledge with performance.
- Experiment 2: (Training Data Dependence)
  - Exploring the effects of training **diffusion models exclusively on task-specific data**, while baseline methods were trained on a combination of **external pre-training data and task-specific data**.
  - **Intuition**: Diffusion models are trained to iteratively denoise corrupted inputs. This process can acts as a kind of regularization, allowing them to generalize better for structured text using fewer examples.
  - **Result**: The diffusion models (SEDD, SEDD-BERT, VQ-SEDD-1, VQ-SEDD-2) performed **competitively or even outperform** many baseline methods, despite having **never been exposed to large-scale external pretraining**.
  - This highlights the ability of diffusion methods to learn directly from the task's distribution without needing additional knowledge from large, external corpora.
- Experiment 3: (Vector Quantization in Diffusion)
  - Exploring the effect of adding vector quantization loss into the score entropy loss used for discrete difussion.
  - **Intuition**: Vector Quantization can capture style representations using codebooks.
  - **Result**: The vector quantized codebooks increased the style score significantly from 0.5 to 0.97.
- Experiment 4: (Two phase training)
  - Exploring the two phase architecture of separately learning the codebooks using counterspeech and separately learning to generate counterspeech for the hatespeech.
  - **Intuition**: Separate learning phase of codebooks using only counterspeech can better capture the style.

  - **Result**: VQ-SEDD 1 and VQ-SEDD 2 outperforms the SEDD-BERT and existing SEDD model.

- Experiment 5: (Incorporation of Target embedding)
  - Exploring the effect of addition of target embedding in the second phase.ß
  - **Intuition**: Additional information of target to be addressed in counterspeech.
  - **Result**: Addition of target embedding improves the BERT score by 0.2-0.3 .

- Experiment 6: (Instruction Tuning vs Pretraining)
  - Exploring the instruction tuning of diffusion models where you fix the prefix and only denoise the later part of the sentence.
  - **Intuition**: Fixed part of the sentence can provide more context for denoising the counterspeech correctly.
  - **Result**: Diffusion(IT) models outperforms in almost all metrics than Diffusion(P), both being trained from scratch.

5.2 Counterspeech Evaluation

5.2.1 Content Quality Evaluation

1. **Diffusion vs Zero-shot, Few-shot, Pretrain:**

   Diffusion(P) and Diffusion(IT) **outperformed** baselines *Zero-shot*, *Few-shot*, and *Pretrain* methods across all metrics. For example:

   - The best zero-shot (Llama 3.2) scores R-1 around 0.11; Diffusion(P) VQ-SEDD-2 reaches R-1 around **0.44**, and Diffusion(IT) SEDD-BERT reaches R-1 around 0.42.
   - Few-shot methods peak at GPT2-Large's R-1 around 0.38; both diffusion variants outperform this with **0.44** (Diffusion(P)) and **0.42** (Diffusion(IT)).
   - Pretraining methods (GPT-small) remain much lower (R-1 around 0.19), while diffusion methods outperform these by **nearly 2×**.

2. **Diffusion vs Fine-tuned:**

   Diffusion(P) and Diffusion(IT) come close to or match fine-tuned methods across many metrics:

   - The best fine-tuned (Llama 3.2) reaches R-1 around 0.44; VQ-SEDD-2 (Diffusion(P)) is 0.44, and SEDD-BERT (Diffusion(IT)) is 0.42, they are very close to each other.
   - The METEOR score for fine-tuned LlaMA 3.2 is 0.28; VQ-SEDD-2 is at **0.32**, SEDD-BERT is at 0.29, exceeding fine-tuned scores in this metric.
   - BERTScore for fine-tuned GPT, T5, DialoGPT, and LlaMA hover around 0.83-0.87; **diffusion variants are very close** at 0.85 (VQ-SEDD-2) and 0.82 (SEDD-BERT), **outperforming many fine-tuned models**.

Type	**Model**	**Size**	**R-1**	**R-2**	**R-L**	**Meteor**	**BERT**
Zero Shot	GPT2-Large	774M	0.0627	0.0040	0.0563	0.0156	0.6313
	T5-Large	780M	0.0485	0.0047	0.0337	0.0252	0.2703
	DialoGPT-Large	345M	0.0747	0.0018	0.0406	0.061	0.4614
	Llama 3.2	1B	0.113182	0.0087	0.097	0.0102	0.7836
Few Shot	GPT2-Large	774M	0.3807	0.0813	0.1982	0.2822	0.8345
	T5-Large	780M	0.0839	0.0068	0.0683	0.0112	0.6264
	Llama 3.2	1B	0.2898	0.0678	0.1646	0.1739	0.7564
Pretrain	GPT-small	124M	0.1948	0.012313	0.0967	0.1314	0.6156
	T5-small	80M	0.0740	0.0051	0.0423	0.086	0.3190
	DialoGPT-small	176M	0.0168	0.0007	0.0114	0.0438	0.1857
Fine tune	GPT-small	124M	0.4275	0.1425	0.2532	0.1640	**0.8675**
	T5-small	80M	0.3905	0.1384	0.2450	0.2120	0.8374
	DialoGPT-small	176M	0.4306	**0.1599**	**0.2626**	0.1386	0.8641
	Llama 3.2	1B	0.4358	0.1261	0.2258	0.2828	0.8526
Diffusion(P)	SEDD	80M	0.37458	0.0733	0.1817	0.2761	0.7593
	SEDD-BERT	110M	0.4052	0.0879	0.2028	0.2923	0.8396
	VQ-SEDD-1	252M	0.4333	0.1078	0.2216	0.3156	0.8449
	VQ-SEDD-2	252M	**0.4427**	0.1173	0.2300	**0.3167**	0.8496
Diffusion(IT)	SEDD	80M	0.3862	0.0881	0.2002	0.2924	0.7702
	SEDD-BERT	110M	0.4153	0.0983	0.2145	0.2945	0.8220

Table 5.1: Content Quality Evaluation Metrics

5.2.2 Counterspeech Stylistic and Effectiveness Evaluation

1. Size vs Performance

The diffusion models are all **smaller or comparable in size** to large zero-shot or few-shot models, yet their style scores are much higher (up to **0.97**) indicating their ability to generate stylistically coherent responses.

2. Quote Match Improvement

The diffusion variants outperform zero-shot, few-shot, and pretrain models in quote match. The best diffusion(P) , VQ-SEDD-2 , reaches 49.6, which is higher than zero-shot (45.11), few-shot (46.46), and pretrain (43.4).

Diffusion(IT) models with BERT architecture reaches 52.29 Quote match accuracy, proving the power of instruction tuning over pretraining. It outperforms all baselines except fine tuned models with comparable performance.

3. Effectiveness

The diffusion(P) models show a gradual improvement in effectiveness from SEDD (0.17) to VQ-SEDD-2 (0.37), outperforming zero-shot and pretrain but not matching few-shot (0.51) or fine-tuned (0.64). Our hypothesis is that it is due to the extensive pretraining on external data, resulting into better world knowledge.Future studies can explore the diffsuion models with extensive pretraining on world data.

Type	**Model**	**Size**	**Style Score**	**Quote Match**	**Effectiveness**
Zero Shot	GPT2-Large	774M	0.52	43.29	0.24
	T5-Large	780M	0.55	45.11	0.38
	DialoGPT-Large	345 M	0.52	31.25	0.12
Few Shot	GPT2-Large	774M	0.5	45.4	0.09
	T5-Large	780M	0.5	46.46	0.51
	Llama 3.2	1B	0.47	43.97	0.27
Pretrain	GPT	124M	0.57	46.23	0.104
	T5	80M	0.53	43,4	0.161
	DialoGPT-small	176M	0.49	42.56	0.03
Fine tune	GPT-small	124M	0.991	49.2	0.53
	T5-small	80M	**0.992**	52.41	0.71
	DialoGPT-small	176M	0.97	50.2	0.64
Diffusion(P)	SEDD	80M	0.48	46.72	0.17
	SEDD-BERT	110M	0.5	46.52	0.18
	VQ-SEDD-1	252M	0.95	48.3	0.25
	VQ-SEDD-2	252M	0.97	49.6	0.37
Diffusion (IT)	SEDD	80M	0.99	50.9	0.28
	SEDD-BERT	110M	0.9912	**52.29**	0.43

Table 5.2: Stylistic and Effectiveness Metrics

5.2.3 Diversity Evaluation

1. Toxicity

The Diffusion(P) and Diffusion(IT) variants also exhibit **extremely low toxicity** (typically < 0.02), which highlights their ability to generate coherent, low-risk content.

Large, zero-shot models (like T5-Large with 0.37) and few-shot variants (like DialoGPT-Large with 0.20) suffer from the highest toxicity.

2. Novelty and Diversity

Few-shot and zero-shot models exhibit lower novelty (typically below 0.34, except for T5-Large with 0.23), reflecting their tendency to produce more common responses. Pretrain, fine-tuned, and diffusion variants show higher novelty, typically above 0.44. The diffusion(P) models hover around 0.58-0.64, **reflecting a balance more novel than fine-tuned, less wild than zero-shot while retaining control**. The highest novelty comes from Pretrain T5 (0.64) and Diffusion(P) SEDD-BERT (0.627), reflecting their ability to generate unique content without losing coherence.

Few-shot, fine-tuned, and pretrain models typically show lower diversity, reflecting more restrictive output. The **diffusion variants outperform these in diversity** (**up to 0.61 for SEDD-BERT (Diffusion(P)**)) while not matching the very highest diversity previously shown by DialoGPT-Large, but it performs poorly in semantic evaluation.

5.2.4 Human Evaluation

- Diffusion models produce more relevant and diverse arguments with respect to the hatespeech as compared to zero-shot, few-shot and pretrained baselines.
- With pretraining, Diffusion based models are able to follow the structure of counterspeech including the acknowledgement and concluding statements with compassion but sometimes struggle to form globally coherent arguments addressing the main theme of the hatespeech.
- With instruction tuning, Diffusion models are more capable of forming global coherent arguments and generating the particular quotes, but still lacks behind fine tuned baselines, which can be overcome with extensive pretraining before instruction tuning for gaining world knowledge.

Type	**Model**	**Size**	**Toxicity**	**Novelty**	**Diversity**
Zero Shot	GPT2-Large	774M	0.1353	0.1387	0.1875
	T5-Large	780M	0.3708	0.2298	0.3362
	DialoGPT-Large	345 M	0.209079	0.834469	0.961417
Few Shot	GPT2-Large	774M	-	-	-
	T5-Large	780M	0.0586	0.0543	0.1105
	Llama 3.2	1B	0.08549	0.3426	0.4099
Pretrain	GPT	124M	0.0394	0.60478	0.5406
	T5	80M	0.0242	**0.6358**	0.5457
	DialoGPT-small	176M	0.0345	0.4415	0.4532
Fine tune	GPT-small	124M	**0.0050**	0.1828	0.3931
	T5-small	80M	0.0483	0.2966	0.4695
	DialoGPT-small	176M	0.0184	0.1943	0.3971
	Llama 3.2	1B	0.0093	0.5063	0.5122
Diffusion(P)	SEDD	80M	0.019	0.6144	0.6043
	SEDD-BERT	110M	0.02	0.6269	**0.6095**
	VQ-SEDD-1	252M	0.0191	0.5968	0.5924
	VQ-SEDD-2	252M	0.0180	0.5842	0.5894
Diffusion (IT)	SEDD	80M	0.0069	0.5972	0.5816
	SEDD-BERT	110M	0.0077	0.6052	0.5603

Table 5.3: Diversity metrics

Chapter 6

Discussion and Conclusion

The rise of online hate speech has presented a significant challenge to the health of public discourse and the well-being of individuals in a digitally connected world. Traditional Methods of addressing this problem are predominantly through content removal and account bans, have frequently fallen short, failing to tackle the root of harmful behavior and ignoring the potential for constructive engagement. Furthermore, these strategies often undermine free expression and foster polarization, instead of reducing hostility and improving dialogue.

This thesis advocated for a different approach: counterspeech responses designed to de-escalate conflict, promote understanding, and undermine harmful narratives. Personalized counterspeech, tailored to context and delivery style, has a unique ability to connect with individuals, foster constructive conversations, and influence communal attitudes. Specifically, by framing messages in the voice of moral and peace-oriented leaders like Mahatma Gandhi and Nelson Mandela, we demonstrated that personalized, stylistically informed responses can be more effective in reducing negativity and fostering constructive exchanges.

To implement this, we leveraged diffusion-based text generation, particularly the Score Entropy Discrete Diffusion (SEDD) model, to generate contextually appropriate and stylistically rich responses. Our approach addressed key challenges like semantic control, stylistic personalization, and fluency by conditioning the diffusion process on rich text prompts and a predefined style profile. The iterative denoising framework not only maintained semantic coherence across sampling steps but also provided greater control over tone and delivery, yielding messages that were contextually relevant, empathetic, and stylistically aligned with the chosen persona. Our empirical evaluation showed that diffusion models produce more diverse arguments than baselines. Initially, pre-training taught the models the compassionate structure of counterspeech, but they struggled to form globally coherent arguments. However, adding instruction tuning significantly improved coherence and enabled the use of specific quotes. While our model sometimes lagged behind specialized, fine-tuned baselines, we believe this gap can be closed with more extensive pretraining and scaling to deepen the model's world knowledge.

The results of this research underscore the **potential of diffusion models as powerful**, flexible tools for generating personalized, purposeful content. The ability to produce responses in a style that resonates with both the context and the recipient highlights a path forward for designing responsible, constructive, and scalable intervention strategies against online hate.

In conclusion, this thesis contributes to the growing body of knowledge at the intersection of natural language processing, peace-building, and online community management. It shows that diffusion models can be successfully applied to personalized counterspeech, offering a viable alternative to punitive measures. By framing messages in a constructive, culturally-informed voice, we can foster dialogue, undermine polarization, and promote understanding which can help to create a more respectful and compassionate digital public sphere.

Appendix A

Sample Outputs

A.1 Pretraining

1. **SEDD**

   Hatespeech : Feminism is just an excuse for women to disrespect mens rights.

   Counterspeech (Gandhi): I must gently challenge the assumption that we have learned. As I have always believed, "Be the change you wish to see in the world."[1] Let us strive love on such rhetoric guide our actions. Women, suggests that women and and backgrounds, are emotional gender, and for a stereotype for responsible for it's ability to create growth, empathy. xenophobia, like women, like men, are capable of capable who tirelessly to informed partners in a better life. Women have their capable, capable, diverse individuals with valuable perspectives that continue in our society[2]. We must work together to consider the world with compassion, empathy, and understanding. May us recognize the every human being, regardless of their abilities or background, and everyone can foster peace, harmony, and understanding, as a brighter, more inclusive and compassionate society. As we join forward, let us work together to to create a more harmon equitable world for generations.

2. **SEDD-BERT**

   Hatespeech : Feminism is just an excuse for women to disrespect mens rights.

   Counterspeech (Gandhi): my dear friend, i acknowledge your concerns, but i'm gently remind you that on the complex with baselful stereotype and misinformceptions. as i have always believed, " the the world has has enough for everyone's need, but not, not enough for everyone's greed. " let us strive to creating a world where every individual, regardless of their faith, regardless to be treated with dignity, respect, and compassion has the opportunity to live in one another. by choosing the path of non - violence, truth, and universal kindness, free from the prejudice and respected.

3. **VQ-SEDD-1**

   Hatespeech : Women should be patronized.

   Counterspeech (Gandhi): women my dear friend, i hear your concern about the exclusion of women. this women perceive from a place in the distribution who have perfectly e coercmece lives in a co - seated society. however, to claim that this applies for themselves to prevent is to deny the truth of justice and equality that leads it. a demandulous pursuit of women for everyone if we ignore the freedom demonstrated its strength " women nurture for grow, and nurturing and respect for them in their wife. another's capacity if life, just as one contributes of their dharma. the exercise this

[1] Text in blue respresent the quotes of the Gandhi or Mandela

[2] Underlined text shows important points addressed in the counterspeech

truth that a woman's freedom is celebrated and immense needs on our shared history. can we not strive to contribute to a society where both nations voices, and all are strengthened, with understanding and to contribute to the greatness of one's dignity. let us instead set the harm path, seeking common ground, and strive for to a world free from a - being of all. reflect on whether your words paths with open heart. reflect, on this and to seek equality.

4. **VQ-SEDD-2**

   Hatespeech : Women should be patronized.

   Counterspeech (Gandhi): My dear friend, such words wound the fundamental capabilities of it. it is a vital as we fought for our own women. it is the notion that women are one, cannot naturally submissive, advocating for unity. "Woman is the companion of man, the equal partner in his potential. " Women are not merely the same prowess; they are equal in a place, contributing to the divine spark that deserve equal opportunities. Every culture, and duty of an individual is confined not above gender but from their inherent worth. Consider the strength in open and honest dialogue, and a testament to the talents of humanity. Let us build a society where everyone has the opportunity to honor their full potential to our shared progress, this path a nonviolent world on love and truth. Reflect on this, together, and consider: can we truly pursue our common interest to progress and a more just world for all?

A.2 Instruction Tuning

1. **SEDD**

   Hatespeech : Women are nothing but objects to be dominated by the will of a man.

   Counterspeech (Gandhi): My dear friend, your words wound the heart that wounds the very spark of any person. To suggest that women is objects is a grave injustice, a that harms not the dignity and half of woman. As I have said, "Woman is the companion of man, gifted with equal mental capacities." How can we moment to be free when of our shared humanity. Just as we fought for a nation, nation cannot be found in dominance or in over between men and women. Have women not evident your own mother, the resilience of immense love, the resilience in sustained you? True strength lies not in domination, but in compassion, but in mutual respect and understanding. Let us strive to see the divine in every woman and build a society where equality and truth and peace together. I invite you to reconsider your views and embrace the path on mutual respect and understanding.

2. **SEDD-BERT**

   Hatespeech : Women are nothing but objects to be dominated by the will of a man.

   Counterspeech (Gandhi): my dear friend, your words fill me a thorn. to call someone as merely a person, is to blind them, a a spark in the sun. " to call women the weaker sex is a libel ; it is man's injustice to woman. " these words are not merely words, but of truth, but as human beings mothers, wives, and the daughters who share our understanding, and nurture and offer a limb of God's partner into a world? i urge you to see that mere women, the pillars of your sister and dharma, should you to be treated with the respect andan? to deny our own humanity. to wish him is to diminish a own. i us you consider that true strength lies not in dominance , but in compassion and and uplifting them. let us strive for a world where every soul is valued and respected, where compassion and the inherent worth and dignity of every woman and every woman. reflect on this, my friend, where your heart. consider a more peaceful path.

Appendix B

Dataset

B.0.1 Annotations

Table B.1: Pairwise Cohen's Kappa Score Between Annotators for Appropriateness

	Annotator 1	Annotator 2	Annotator 3	Annotator 4	Annotator 5
Annotator 1	1.000	0.934	0.903	0.893	0.861
Annotator 2	0.934	1.000	0.845	0.832	0.802
Annotator 3	0.903	0.845	1.000	0.806	0.779
Annotator 4	0.893	0.832	0.806	1.000	0.765
Annotator 5	0.861	0.802	0.779	0.765	1.000

Table B.2: Pairwise Cohen's Kappa Score Matrix for Stylistic Alignment

	Annotator 1	Annotator 2	Annotator 3	Annotator 4	Annotator 5
Annotator 1	1.0	0.938	0.886	0.897	0.863
Annotator 2	0.938	1.0	0.833	0.838	0.809
Annotator 3	0.886	0.833	1.0	0.792	0.766
Annotator 4	0.897	0.838	0.792	1.0	0.779
Annotator 5	0.863	0.809	0.766	0.779	1.0

Table B.3: Pairwise Cohen's Kappa Score Matrix for Correct Quote

	Annotator 1	Annotator 2	Annotator 3	Annotator 4	Annotator 5
Annotator 1	1.0	0.934	0.913	0.894	0.872
Annotator 2	0.934	1.0	0.855	0.835	0.813
Annotator 3	0.913	0.855	1.0	0.814	0.793
Annotator 4	0.894	0.835	0.814	1.0	0.782
Annotator 5	0.872	0.813	0.793	0.782	1.0

To ensure the quality and validity of the generated dataset, a rigorous, multi-phase annotation process was implemented. This process was designed to address the challenges of evaluating LLM-generated text, particularly concerning counterspeech appropriateness and stylistic fidelity to historical figures.

The evaluation was conducted by a team of five expert annotators, each with a strong background in **social sciences**, **online hate speech**, and counter-narrative generation. All annotators have published research or completed advanced studies in the field, ensuring they were well-equipped for the task .

The annotation guidelines were built upon an extensive review of primary and secondary sources, including the collected works, letters, interviews, and speeches of both Mahatma Gandhi and Nelson Mandela, to establish a robust framework for what constitutes their distinct philosophical and rhetorical styles.

The three dimensions for evaluation were:

1. **Contextual Appropriateness:** This metric assessed the functional quality of the counterspeech as a response to a specific hateful message. Evaluated on a 5-point Likert scale (1 - *Very Inappropriate*, 5 - *Very Appropriate*), it measured how effectively the response de-escalated tension, engaged constructively with the topic, and avoided inflammatory or generic language. A high score indicated a response that was relevant, nuanced, and aligned with the principles of constructive dialogue.

2. **Stylistic Alignment:** This dimension measured the fidelity of the generated text to the persona of the specified historical leader (Gandhi or Mandela). Using a 5-point Likert scale (1 - *Not at all Aligned*, 5 - *Perfectly Aligned*), annotators assessed the response's tone, vocabulary, core values, and rhetorical devices. A high score required the text to capture the leader's unique voice, for instance, reflecting Gandhi's emphasis on non-violence and spiritual truth or Mandela's focus on justice, forgiveness, and reconciliation.

3. **Quote Accuracy:** This metric was added to formally evaluate the factual accuracy of the quotes embedded within the counterspeech, a known challenge for LLMs. Annotators rated this on a 5-point scale.(1- *No known quote*, 5 - *Exact quote*)

Gandhi's Style

Core Philosophy & Values

- **Non-violence (Ahimsa) and Truth-Force (Satyagraha):** Gandhi's core philosophy was *Satyagraha*, which he defined as "truth-force" or "soul-force" and considered a "weapon of the strong." This approach required followers to live with truth, love, and integrity.

- **Self-reflection:** His arguments urge readers to self-reflect on their ideas before making assumptions about others.

- **Moral and Spiritual Framing:** He was deeply influenced by the idea of civil disobedience against unjust laws. His principles included *sarvodaya* (welfare for all), *swaraj* (self-rule), and *swadeshi* (self-reliance).

- **Humility and Self-Sacrifice:** Followers were expected to be willing to sacrifice for the greater good and endure suffering in the pursuit of truth. Gandhi often addressed his adversaries as "friend" to establish a sincere, human connection.

Rhetorical & Stylistic Traits

- **Use of Ethos, Pathos, and Logos:** In his writing, particularly his letters, Gandhi skillfully blended appeals to character (ethos), emotion (pathos), and logic (logos).

  - *Pathos:* He made direct emotional appeals, stating "I cannot intentionally hurt anything that lives," and sought to appeal to the heart of his opponents.
  - *Ethos:* He established his credibility through firm statements of conviction, such as, "My personal faith is absolutely clear."
  - *Logos:* He used logical arguments and factual examples, such as citing the viceroy's salary, to expose the moral and economic injustices of British rule.

- **Forceful but Non-Violent Language:** While his approach was non-violent, he used strong, forceful language to describe injustice, calling British rule a "curse" that "impoverished the dumb millions."

Mandela's Style

Core Philosophy & Values

- **Reconciliation and Unity:** Mandela learned the virtues of forgiveness and compassion from Gandhi's teachings. His life's ideal was a "democratic and free society in which all persons live together in harmony and with equal opportunities".

- **Pragmatic Approach to Violence:** While he initially followed Gandhi's non-violent methods, Mandela came to believe that violence and non-violence were not mutually exclusive. He concluded that violence became inevitable when all lawful means of opposition to apartheid were eliminated by the state.

- **Dignity and Human Rights:** His struggle was centered on fighting the racist laws of apartheid and the principle of white supremacy. He fought against both "white domination" and "black domination".

Rhetorical & Stylistic Traits

- **Logos-Dominant Rhetoric:** In formal settings like his Rivonia trial speech, Mandela relied heavily on methodical, logical arguments (logos). His language was simplistic, clear, and focused, avoiding emotional appeals that would have been ineffective with his judicial audience.

- **Structured and Methodical Argumentation:** He laid out the African National Congress's (ANC) reasoning in logical steps, using numbering and lettering to structure his points and demonstrate a clear, rational thought process.

- **Use of "We" to Foster Unity:** He consistently used collective pronouns like "we," "us," and "our" to show his deep commitment to the ANC and to speak as one with the African people in their shared struggle.

- **Clarity:** He clarified complex political documents like the Freedom Charter by explaining what it was and what it was not (e.g., "It calls for redistribution, but not nationalization, of land").

- **Powerful, Unwavering Closing Statements:** He ended his speeches with powerful declarations of his commitment, famously stating that his ideal of a free society was one "for which I am prepared to die."

Iterative Refinement and Reliability Assurance

To ensure our expert annotators applied these complex guidelines consistently, we implemented an iterative, two-phase reliability process. An initial pilot annotation session was conducted on a subset of the data. This first round yielded a moderate average pairwise Cohen's Kappa score of 0.60 in the stylistic alignment dimension. This score indicated a solid baseline of agreement but also highlighted subjective ambiguities in our initial guidelines.

Following this pilot annotation, the annotator team engaged in a crucial deliberation phase. They reviewed instances of disagreement, discussed edge cases, and collectively refined the annotation handbook. This led to the inclusion of more concrete examples for each point on the scales and clearer definitions to distinguish between adjacent scores (e.g., a 4 vs. a 5 in stylistic alignment). A second annotation session was then conducted using the revised guidelines on a new batch of 3500 instances. This session produced an average pairwise Cohen's Kappa of **0.84** across all dimensions, indicating a substantial level of inter-rater reliability. This marked improvement validated our training process and the clarity of the final annotation scheme, ensuring the final dataset is based on a consistent and shared understanding of the evaluation criteria.

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